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Jnana Sangama, Belagavi – 590018



AI Based Stock Market Prediction using LSTM

Submitted in partial fulfillment of the requirements as a part of the curriculum,
Bachelors of Engineering in Artificial Intelligence and Machine Learning

Submitted by

SANJANA R 1CR24AI116

K LAKSHMI DEEKSHITHA 1CR24AI055

SAMIKSHA YADAV 1CR24AI112

Under the Guidance of

Dr. Shyam P Joy

Head Of Department

Department of Artificial Intelligence and Machine Learning



Department of Artificial Intelligence and Machine Learning
CMR INSTITUTE OF TECHNOLOGY

132, AECS Layout, Kundalahalli, ITPL Main Rd, Bengaluru – 560037



CERTIFICATE

Certified that the Mini Project entitled “AI Based Stock Market Prediction using LSTM” is presented by

Sanjana R **1CR24AI116**

K Lakshmi Deekshitha **1CR24AI055**

Samiksha Yadav **1CR24AI112**

bonafide student(s) of **CMR Institute of Technology** in partial fulfillment of the requirements as a part of the curriculum, **Bachelors of Engineering in Artificial Intelligence and Machine Learning**, of **Visvesvaraya Technological University, Belagavi** during the year **2024-28**. It is certified that all correction/suggestion indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The Entrepreneurial Project report has been approved as it satisfies the academic requirements in respect of the Technical Seminar prescribed for the bachelor of engineering degree.

(Dr.Venkata Gopal)

(Dr. Shyam P Joy)

Signature of the Guide

Signature of the HOD

DECLARATION

We, the undersigned, students of **CMR Institute of Technology**, hereby declare that this mini project report titled

“AI Based Stock Market Prediction using LSTM”

is a record of our original work done under the supervision of Dr. Shyam P Joy, Head of Department, Department of Artificial Intelligence and Machine Learning. It has not formed the basis of award of any other degree, in this or any other Institution or University. In keeping with the ethical practice in reporting scientific information, due acknowledgements have been made wherever the findings of others have been cited.

1. Sanjana R (USN: 1CR24AI116)(Signature)
2. K Lakshmi Deekshitha (USN: 1CR24AI055)(Signature)
3. Samiksha Yadav (USN: 1CR24AI112)(Signature)

ABSTRACT

This mini project presents an AI-based stock market prediction system using Long Short-Term Memory (LSTM), a deep learning model suitable for time-series data. The system uses historical stock data containing Open, High, Low, Close and Volume values. The dataset is collected using stock market data sources and stored in CSV format for stable access. The data is preprocessed using normalization and converted into 30-day time sequences. These sequences are used to train the LSTM model to predict the next day closing price of a stock. The trained model is saved and later used for prediction without retraining every time. A simple dashboard interface is developed using Streamlit and Plotly to display stock charts, predicted values and actual versus predicted price comparison. The project helps in understanding how deep learning can be applied to financial time-series forecasting, while also showing the limitations of using only historical price data.

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It is my proud privilege and duty to acknowledge the kind of help and guidance received from several people in preparation of this report. Apart from my own, the success of this report depends largely on the encouragement and guidelines of many others. It would have not been possible to prepare this report in this form without their valuable help, co-operation and guidance.

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1. INTRODUCTION

1.1 Overview

Stock market prediction is an important application area of Artificial Intelligence and Machine Learning. Stock prices change every day based on many factors such as company performance, investor sentiment, market trends, economic conditions and global events. Because of this, predicting stock prices is difficult and cannot be done accurately using only simple manual observation.

In this project, a deep learning approach is used to predict stock prices using historical data. The system takes previous stock values such as Open, High, Low, Close and Volume and learns patterns from them. Since stock data is sequential in nature, an LSTM model is used because it can remember previous time steps and learn trends from a sequence of days.

The project also includes a dashboard-style interface where users can view stock data, graphs and predicted values. This makes the system easier to understand and more useful than only showing terminal output.

1.2 Problem Statement

Stock prices are highly volatile and difficult to predict because they depend on multiple changing factors. Traditional methods may not capture complex patterns in historical stock data effectively. Investors, students and analysts often need tools that can study past stock behavior and provide an estimated future trend. If such prediction support is not available, stock analysis becomes more time-consuming and less systematic. This project aims to build a deep learning based system that predicts the next day closing price using previous stock data.

1.3 Objectives

- To collect historical stock market data for a selected company.
- To preprocess the dataset by cleaning, scaling and converting it into time-series sequences.
- To design and train an LSTM model for next-day stock closing price prediction.
- To evaluate the model by comparing predicted prices with actual historical prices.
- To develop a simple dashboard that displays stock charts and prediction results.
- To understand the practical use of deep learning in financial forecasting.

1.4 Scope of the Project

This project focuses on predicting the next day closing price of a selected stock using historical numerical data. The current system uses features such as Open, High, Low, Close and Volume values. It includes data loading, preprocessing, LSTM model training, prediction and basic dashboard visualization.

The project does not guarantee real-world trading accuracy and does not consider external factors such as news, company announcements, government policies or social media sentiment. It is intended as an academic deep learning project for understanding time-series prediction.

2. LITERATURE SURVEY

- Yahoo Finance and yfinance are commonly used for collecting historical stock market data such as Open, High, Low, Close and Volume. These sources make it easier to build machine learning models using real market data. However, they only provide data and do not perform prediction by themselves.
- Recurrent Neural Networks (RNNs) are used for sequential data, but they can face difficulty in remembering long-term dependencies. LSTM models improve this by using memory cells and gates, making them more suitable for time-series forecasting tasks like stock prediction.
- TensorFlow and Keras provide tools for building and training deep learning models. Keras simplifies model creation by providing easy layer-based APIs, while TensorFlow performs the backend computation and optimization.
- Streamlit and Plotly are useful for creating simple interactive data applications. Streamlit helps convert Python code into a web application, while Plotly provides interactive graphs such as line charts and candlestick charts. The limitation is that the quality of prediction still depends on the trained model and dataset.

3. METHODOLOGY

3.1 SYSTEM DESIGN

The system is designed as a pipeline where stock data is collected, preprocessed, passed through an LSTM model and displayed through a dashboard. The design separates the project into different modules such as data loading, preprocessing, model building, training, prediction and visualization. This makes the project easier to maintain and understand.

3.1.1 System Requirements

Functional Requirements

- The system shall load historical stock data from a CSV file or online stock source.
- The system shall preprocess the data by scaling numerical values.
- The system shall create 30-day time sequences for LSTM input.
- The system shall train an LSTM model using historical stock data.
- The system shall save the trained model for later prediction.
- The user should be able to view predicted and actual stock prices through graphs.

Non-Functional Requirements

- The system should be simple and easy to use for students and evaluators.
- The model should run on a normal laptop CPU without requiring a high-end GPU.
- The code should be modular and divided into separate files for better readability.
- The dashboard should display results clearly using graphs.
- The system should be extendable to support more stocks in the future.

Hardware and Software Requirements

Category	Development
OS	Ubuntu/Linux or Windows 10/11
Processor	Intel Core i5 or above
GPU	Not mandatory; CPU execution used
RAM	8 GB RAM or above recommended
Software	Python 3.x, VS Code, Git
Libraries	TensorFlow, Keras, NumPy, Pandas, Scikit-learn, yfinance, Streamlit, Plotly

3.1.2 System Architecture / High-Level Design (HLD)

The system follows a layered architecture. The data layer stores historical stock data in CSV format. The preprocessing layer normalizes the data and creates time-series sequences. The model layer contains the LSTM neural network used for training and prediction. The application layer uses Streamlit to display results, while Plotly is used for interactive charts.

System Flow
Stock Data Source / CSV
Data Loader (data_loader.py)
Preprocessing: scaling + 30-day sequences
LSTM Model Training (model.py + train.py)

Saved Model (stock_model.h5)
Prediction Module (predict.py)
Streamlit Dashboard with Plotly Graphs (app.py)

Figure 3.1: System Architecture Diagram

3.1.3 Database Design (ER Diagram / Schema)

Not Applicable. This project does not use a traditional database. The stock data is stored in CSV format as stock_data.csv. The trained model is stored as stock_model.h5. During execution, the CSV file is loaded using Pandas and processed directly in memory.

Figure 3.3: Database Design - Not Applicable

3.1.4 Module Description

No.	Module Name	Description
1	Data Loading	Downloads or reads historical stock data and returns it as a Pandas DataFrame.
2	Data Preprocessing	Selects Open, High, Low, Close and Volume columns, scales data and creates 30-day sequences.
3	LSTM Model	Builds a deep learning model that learns time-series patterns from stock data.
4	Training and Saving	Trains the model, evaluates loss and saves the trained model as stock_model.h5.
5	Dashboard and Visualization	Displays stock charts, predicted prices and actual versus predicted comparison using Streamlit and Plotly.

3.1.5 Dataset Description

The dataset used in this project contains historical stock market data for Apple Inc. with ticker symbol AAPL. The data contains approximately 1006 trading-day records from 2020 to 2023. Each record represents one trading day.

The main features used are Open, High, Low, Close and Volume. The target variable is the next day Close price. The data is numerical and time-series based. Since the stock market is closed on weekends and holidays, the number of rows is lower than the total calendar days in the selected period.

3.1.6 Model(s) Used

The final model used in this project is an LSTM neural network. LSTM is selected because stock prices are time-dependent and previous days can influence future price movement. The model uses 30 previous trading days as input and predicts the next day closing price.

The architecture includes two LSTM layers, Dropout layers to reduce overfitting and a Dense output layer that produces one predicted value. The model is trained using the Adam optimizer and Mean Squared Error loss function, which is suitable for regression problems.

3.2 IMPLEMENTATION

The project is implemented using Python. Each major part of the system is separated into individual files. `data_loader.py` is used for data collection, `preprocess.py` is used for scaling and sequence creation, `model.py` defines the LSTM model, `train.py` trains and saves the model, `predict.py` loads the saved model and generates predictions, and `app.py` is used for the dashboard interface.

3.2.1 Technology Stack

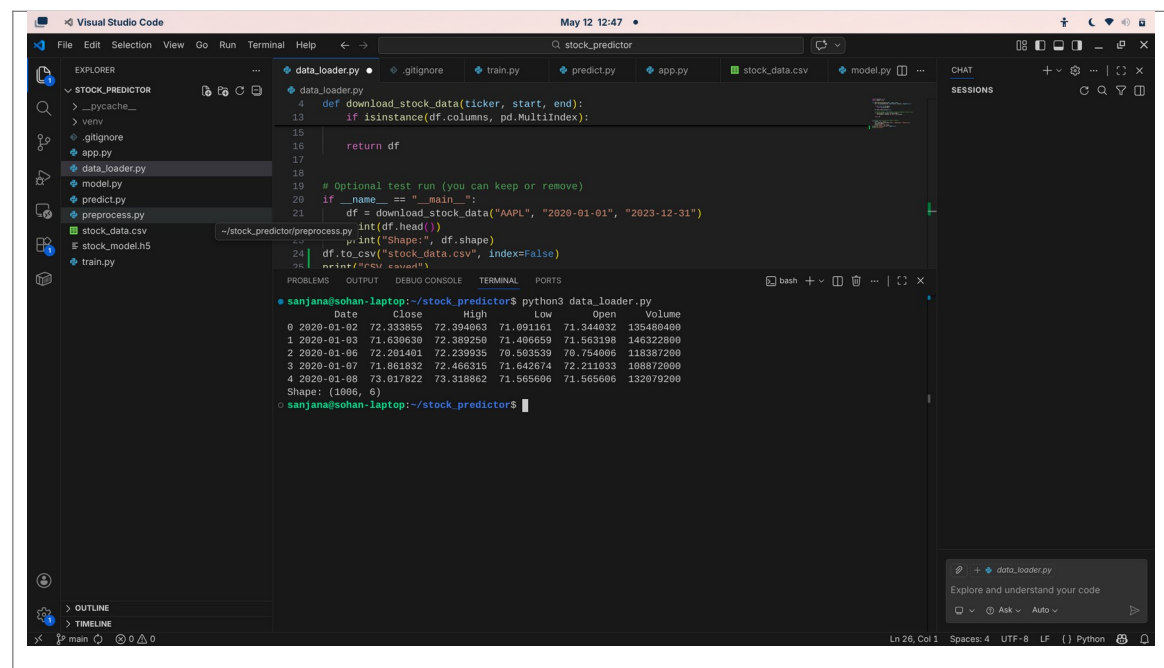
Layer	Technology / Tool	Reason for Choice
Frontend / Dashboard	Streamlit	Simple Python-based dashboard creation with less frontend coding.
Visualization	Plotly	Provides interactive stock-style charts and actual versus predicted graphs.
Data Handling	Pandas, NumPy	Useful for reading CSV files and numerical processing.
ML/AI Library	TensorFlow / Keras	Used to build, train and save the LSTM deep learning model.
Deployment	Local Streamlit App	Suitable for college demonstration and local testing.

3.2.2 Implementation of Key Modules

The project implementation is divided into modules so that each part has a clear responsibility. This makes debugging and future improvement easier. The main modules are data loading, preprocessing, model training, prediction and dashboard visualization.

Module 1: Data Loading and Preprocessing

The data loading module downloads or reads stock data and stores it as a Pandas DataFrame. The preprocessing module selects numerical features such as Open, High, Low, Close and Volume. The values are normalized using MinMaxScaler and converted into 30-day sequences for the LSTM model. Each sequence is used to predict the next day closing price.



```
def download_stock_data(ticker, start, end):
    if isinstance(df.columns, pd.MultiIndex):
        return df

    # Optional test run (you can keep or remove)
    if __name__ == "__main__":
        df = download_stock_data("AAPL", "2020-01-01", "2023-12-31")
        print(df.head())
        print(df.shape)
        df.to_csv("stock_data.csv", index=False)
        print("CSV saved")

sanjana@sohan-laptop:~/stock_predictor$ python3 data_loader.py
Date      Close      High      Low      Open      Volume
0 2020-01-02  72.333855  72.394063  71.091161  71.344032  135488400
1 2020-01-03  71.630630  72.389250  71.406659  71.563198  146322800
2 2020-01-06  72.201401  72.239935  70.503539  70.754006  118387200
3 2020-01-07  71.861832  72.466315  71.642074  72.211033  108872000
4 2020-01-08  73.017822  73.318862  71.565606  71.565606  132079200
Shape: (1006, 6)
```

Figure 4.1: Screenshot placeholder for module implementation

Module 2: LSTM Training, Prediction and Dashboard

The model module builds the LSTM network using TensorFlow and Keras. train.py trains the model using prepared sequences and saves the trained model as stock_model.h5. predict.py loads this saved model and compares predicted prices with actual values. app.py uses Streamlit and Plotly to display stock charts and prediction results in a dashboard format.

3.2.3 Data Preprocessing

The preprocessing step begins by selecting the required numeric columns: Open, High, Low, Close and Volume. These values are scaled using MinMaxScaler so that all features fall within a common range. This helps the neural network train more smoothly.

The scaled data is then converted into sequences using a sliding window method. In this project, the previous 30 trading days are used as input and the next day Close value is used as the target output. This creates 976 input sequences from the dataset.

3.2.4 Feature Engineering

The main features used are Open, High, Low, Close and Volume. These features are directly taken from the stock market dataset because they describe the daily price movement and trading activity. No additional technical indicators such as RSI or MACD are currently used. The main feature engineering step is the creation of 30-day sequential windows for LSTM input.

3.2.5 Model Training

The LSTM model is trained on the prepared 30-day sequences. The model uses two LSTM layers, Dropout layers and a Dense output layer. Training was performed on a CPU environment. The model was trained for 10 epochs with a batch size of 32.

Mean Squared Error is used as the loss function because the project predicts a continuous numerical value. Adam optimizer is used because it is efficient and commonly used in deep learning models. After training, the model is saved as stock_model.h5.

3.2.6 Model Evaluation

The model is evaluated by comparing the predicted closing prices with the actual historical closing prices. Since this is a regression problem, loss is measured using Mean Squared Error. During training, the model achieved a test loss of approximately 0.0028 on scaled data.

The predicted values are inverse-transformed back to original stock price scale and compared with actual values. A graph of actual versus predicted prices is used to visually analyze model performance.

3.2.7 Comparison of Models

Only one final model, LSTM, is implemented in the current version of the project. LSTM was chosen because it is more suitable for sequential time-series data compared to a simple dense neural network. Future versions can compare LSTM with models such as Linear Regression, Random Forest, GRU and ARIMA.

3.2.8 Model Deployment

The trained model is integrated into a Streamlit dashboard. The dashboard loads the saved model and stock dataset, prepares the input data, generates predictions and displays the results using Plotly graphs. This allows users to view the stock chart and prediction output in a web-style interface instead of only terminal output.

4. RESULTS & DISCUSSIONS

5.1 Testing Methodology

The system was tested using both unit testing and integration testing. Individual modules such as data loading, preprocessing, model training and prediction were tested separately. Integration testing was performed by running train.py and predict.py to check the complete workflow from data loading to prediction. The dashboard was tested by running app.py through Streamlit and checking whether graphs and predicted values are displayed correctly.

5.2 Screenshots of Results

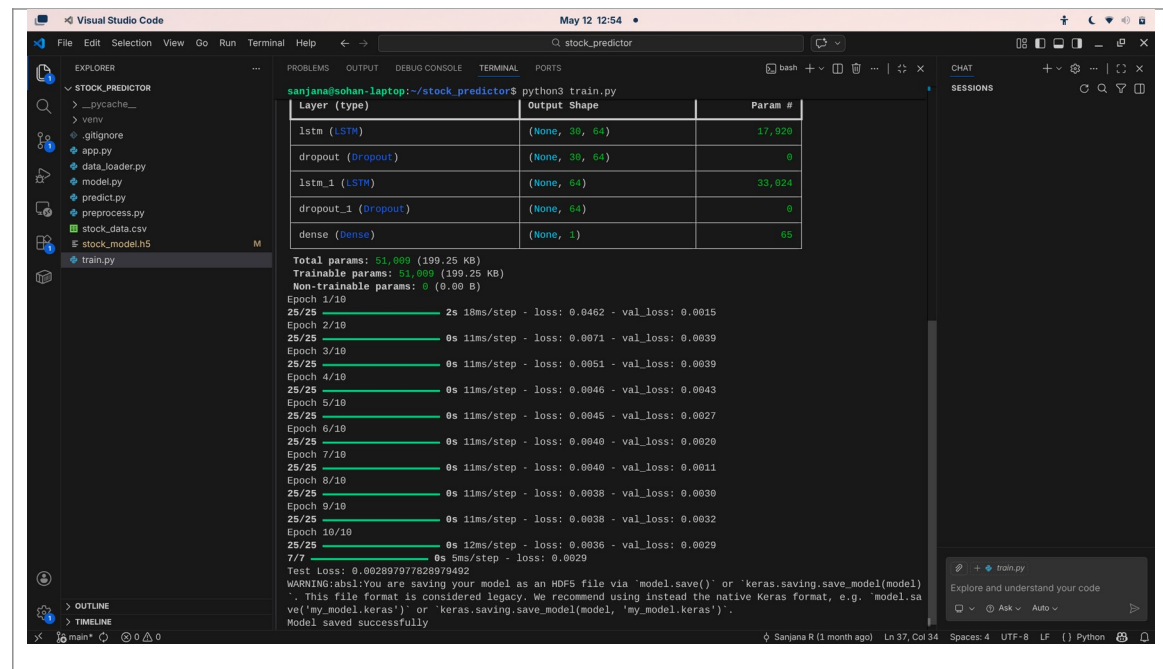


Figure 5.1: Training of the LSTM model using historical stock market data. The model was trained over multiple epochs and the loss values decreased progressively, indicating successful learning.

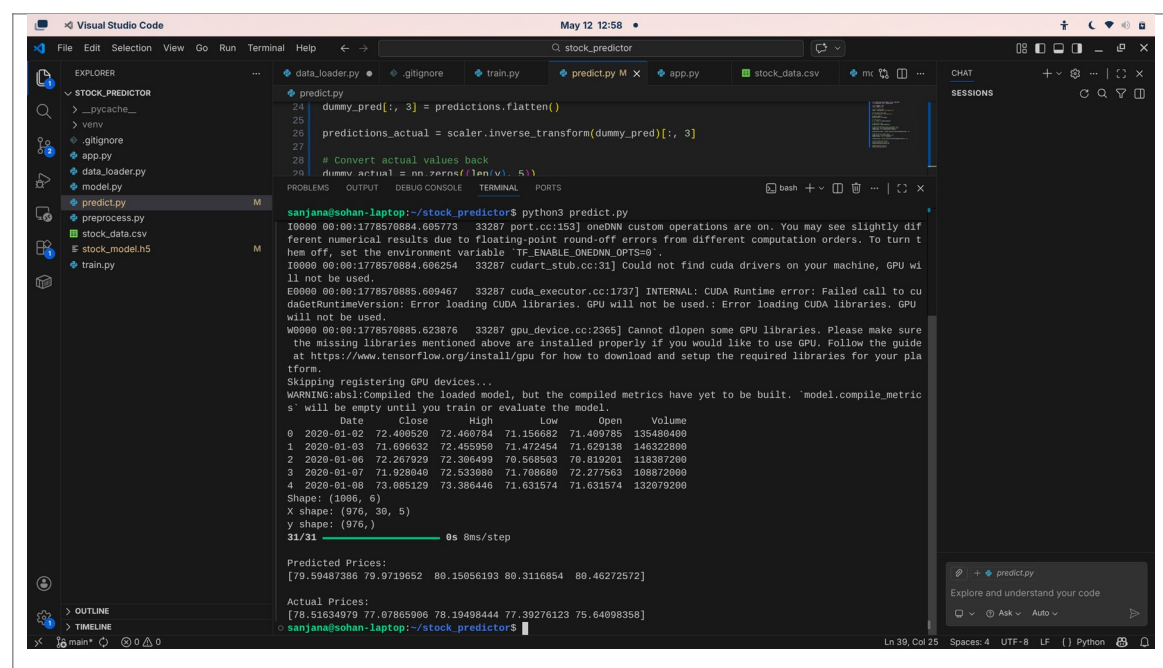


Figure 5.2: Prediction results generated by the trained LSTM model. The predicted stock prices were compared with actual historical prices for evaluation of model performance.

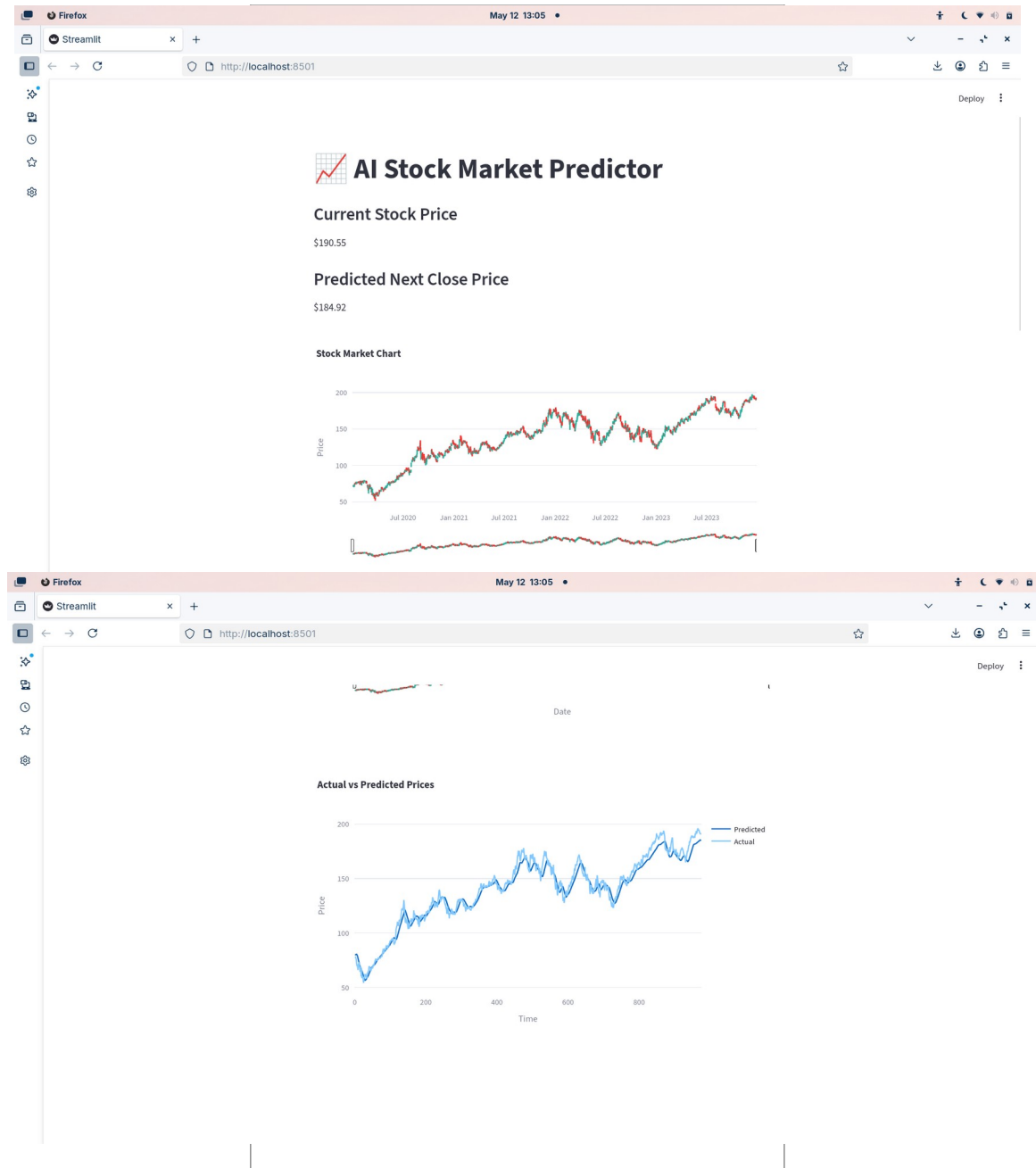


Figure 5.3: Web-based stock market prediction dashboard developed using Streamlit and Plotly. The dashboard displays stock market visualizations and AI-generated stock price predictions.

5.3 Performance Evaluation

The model performance was evaluated using Mean Squared Error loss. The trained model produced a test loss of around 0.0028 on scaled data. Sample predicted prices were close to actual prices, showing that the model learned general stock movement patterns. However, exact prediction is difficult because the stock market is affected by external factors not included in the dataset.

5.4 Discussion

The project successfully demonstrates how an LSTM model can be used for stock price prediction. The model takes previous 30 trading days as input and predicts the next day closing price. The results show that the predicted curve generally follows the actual price movement.

The project also shows the importance of preprocessing in deep learning. Normalization and sequence creation are necessary before feeding data into the LSTM model. The dashboard improves the usability of the system by presenting results through charts and predicted values.

5. CONCLUSION

5.1 Conclusion

This project developed an AI-based stock market prediction system using an LSTM neural network. The system collects stock data, preprocesses it, trains a deep learning model and predicts future closing prices based on previous 30-day patterns. The model was able to generate predictions and compare them with actual historical prices.

The project met its main objectives by implementing data preprocessing, model training, prediction and dashboard visualization. It helped in understanding practical concepts such as time-series data, normalization, LSTM networks, loss functions, optimizers and model evaluation.

5.2 Limitations

- The current model mainly uses historical numerical stock data and does not consider news, sentiment or company fundamentals.
- The model is trained for one selected stock at a time, such as AAPL.
- Stock market prediction cannot be perfectly accurate because market behavior is highly uncertain.
- The current system is suitable for academic demonstration and should not be used as financial advice.

5.3 Future Scope

- Add support for multiple stock tickers through the dashboard.
- Include technical indicators such as moving average, RSI and MACD.
- Add news sentiment analysis to improve prediction context.
- Compare LSTM with other models such as GRU, ARIMA and Random Forest.
- Deploy the dashboard online so that users can access it from a browser.

6. REFERENCES

1. TensorFlow Documentation, TensorFlow and Keras API Guides.
2. Keras Documentation, Sequential Models and LSTM Layers.
3. Yahoo Finance / yfinance documentation for historical stock market data.
4. Streamlit Documentation for Python web applications.
5. Plotly Documentation for interactive financial charts.
6. S. Hochreiter and J. Schmidhuber, "Long Short-Term Memory," Neural Computation, 1997.

APPENDIX A: SOURCE CODE

The complete project source code is available in the GitHub repository below. Only important module links are mentioned here, and the full files can be viewed directly in the repository.

GitHub Repository: <https://github.com/sanjana1aiml24-svg/stock-prediction>

A.1 Module / File Name: app.py

<https://github.com/sanjana1aiml24-svg/stock-prediction/blob/main/app.py>

A.2 Module / File Name: train.py, model.py and preprocess.py

<https://github.com/sanjana1aiml24-svg/stock-prediction/tree/main>